The Standard Model's Structure from $(\mathbb{S}, \tau)$: Gauge Group, Exact Charges, a Native Chirality Mechanism, and the Proven Boundary of $J_3(\mathbb{O})$

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Revision 1.2 — closed. (Rev. 1.2: the quark-column price now shown with its own inline arithmetic, completing the equations-carry-conventions policy per the reviewer's completeness note. Rev. 1.1: collision-proofed the §9 lattice equations per review; disambiguated Theorem 2's branching notation. Hostile read completed on the reviewing side with no mathematical corrections — one presentation fix, one notation clause; author-side audit 40/40 load-bearing values against the verified ledger; all reference titles gate-confirmed from source files.) Supersedes *The Standard Model Gauge Group from* $J_3(\mathbb{O})$ (final Rev. 1.2), *Chirality from* $\mathbb{O}e_8$ (final Rev. 2), and the bridge and Phase-2 content of the connection notes, which remain the detailed working record. Every census number in this paper is verified by **two independent implementations** (separate octonion arithmetic, separate derivation-system assemblies, separate stabilizer extractions, with the lattice results additionally in exact symbolic arithmetic); every structural claim is either proven in closed form or checked to machine precision in both, with the complete checkpoint and deviations record in the development notes. Process history lives there, not here.

Abstract. We derive, from two objects this corpus's earlier work made load-bearing for unrelated reasons — the mediator direction e fixing $\mathrm{Im}(\mathbb{O})$'s complex structure, and the single idempotent A it determines in $J_3(\mathbb{O})$ — the following, each verified twice: the Standard Model gauge group, with color as the mediator's stabilizer in $\mathfrak{g}_2$ (identical in construction to the published exceptional-Jordan program's) and $\mathfrak{su}(2)_{\text{weak}} \oplus \mathfrak{u}(1)_Y$ as A's stabilizer inside the second $\mathfrak{su}(3) \subset \mathfrak{f}_4$; three branching theorems fixing every hypercharge ratio in the 26-dimensional matter census with zero free parameters, one confirmed against exact pre-registered predictions; electric charge $Q := T_3 + Y$ landing every one of 26 states on a Standard Model value with two hand-set conventions; a chirality mechanism native to $\mathbb{S}$'s own τ-odd half, which carries an exact $\mathrm{Cl}(0,8)$ whose evenness supplies a real chirality operator with full, proven $\mathfrak{so}(8)$ triality; a bridge theorem identifying the gauge side's Peirce-$\frac{1}{2}$ module with the $\mathbb{O}e_8$ Clifford module — same module, same even structure, opposite real signature — under an explicit dictionary; the identity $T_3 = \lambda J$: weak isospin's Cartan generator *is* the transported chirality operator, one axis of a quaternionic structure $J_{\frac{1}{2}} \cong \mathbb{H}^4$ on which $\mathfrak{su}(2)_{\text{weak}}$ acts as $\mathfrak{sp}(1)$; the full rank-6 torus of $\mathfrak{e}_6$ with **every generator a named function of the mediator and the idempotent frame**; the complexified 27-census — twelve conjugate pairs governed by a proven pairing theorem plus the idempotent frame itself as three self-conjugate states; and the lattice-point results: u_R's charge reachable at $\alpha = \frac{3}{2}$ exactly, at a proven price of two ratio units paid by whichever sector is not held fixed, with e_R's slot structurally absent. These boundaries are theorems, not gaps: we prove the embedding $G_{SM} \hookrightarrow E_6$ selected by this construction is **inequivalent** to the $SO(10)$-GUT embedding — three computed witnesses, culminating in a conjugacy separation by branching multisets inside one and the same $\mathfrak{so}(10)$ — which locates exactly where the published program's further extensions become necessary rather than optional. What this corpus adds to that program: a derivation of the octonionic seed's *selection* (the separately-proven enumeration theorem), the mediator as the single shared origin of color and complexification, and the boundary theorems above.

1. Introduction

Dubois-Violette and Todorov showed the Standard Model gauge group arises as the intersection of two maximal subgroups of $F_4 = \mathrm{Aut}(J_3(\mathbb{O}))$; Boyle gave the idempotent-stabilizer form; Baez and Schwahn generalized the intersection theorem to arbitrary suitable Jordan subalgebra pairs; Krasnov gave the $\mathrm{Spin}(9)$ characterization. This paper reports an independent arrival at that structure by a route the program does not use — the mediator and idempotent of this corpus's own quantization program — and then continues past it: to the chirality mechanism $\mathbb{S}$'s own split supplies, to the theorem identifying that mechanism's carrier with the weak sector's own matter module, and to the exact boundary of what $J_3(\mathbb{O})$ and its complexification contain.

Two standing structural facts frame everything (proofs in the notes; both elementary): neither $\mathbb{S}$ nor $J_3(\mathbb{O})$ embeds in the other (dimension in one direction; commutativity in the other), and $J_3(\mathbb{O})$ is adopted from the literature, not derived from $\mathbb{O}$ — the bridge between the two constructions is at the level of symmetries, $\mathfrak{g}_2 \subset \mathfrak{f}_4$, and, as §7 proves, at the level of one shared module.

Conventions. $\langle \cdot, \cdot \rangle$ is the trace form $\mathrm{tr}(x \circ y)$ on $J_3(\mathbb{O})$, the canonical metric used throughout; degenerate spectra are interrogated only with canonical (Killing or trace) metrics. Hypercharge-type values are quoted in **ratio units** (the colored weak-doublet's $|Y|$ as unit) unless marked raw; every reported non-invariant scalar carries its unit system. The mediator is written e; $A := (0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}e, 0, 0)$.

2. Color: the mediator's $\mathfrak{su}(3)$

[FACT, proven] $\mathfrak{g}_2 = \mathrm{Der}(\mathbb{O})$ is 14-dimensional; the mediator's stabilizer within it is exactly 8-dimensional, acting on the six non-mediator directions of $\mathrm{Im}(\mathbb{O})$ as the genuine fundamental of $\mathfrak{su}(3)$ (irreducible by Schur, commutant dimension 1) under the mediator's own complex structure $J = L_e$. Embedded entrywise in $\mathfrak{f}_4 = \mathrm{Der}(J_3(\mathbb{O}))$ (verified a derivation to machine precision in both implementations), this is, by construction, the published program's color $\mathfrak{su}(3)$: the stabilizer, within G_2, of a fixed unit imaginary octonion.

3. The second $\mathfrak{su}(3)$, $\mathfrak{e}_{6(-26)}$, and electroweak breaking

[FACT, proven] $\mathfrak{f}_4$ is 52-dimensional (null space of the 10206×729 derivation system; both implementations). The centralizer of color within $\mathfrak{f}_4$ is exactly 8-dimensional, closed, with negative-definite Killing form — hence a second, independent $\mathfrak{su}(3)$, the electroweak $\mathfrak{su}(3)$.

[FACT, proven] Adjoining $L(J_0)$, the traceless Jordan multiplications, to all of $\mathfrak{f}_4$ gives $\mathfrak{e}_6$: the closure is by two classical identities, $[D, L_x] = L_{D(x)}$ and $[L_x, L_y] \in \mathrm{Der}$, each stated and machine-confirmed; the Killing signature is exactly -26 (26 positive on $L(J_0)$, 52 negative on $\mathfrak{f}_4$) — the noncompact form $\mathfrak{e}_{6(-26)}$ with maximal compact $\mathfrak{f}_4$. The compact form $\mathfrak{f}_4 \oplus i\, L(J_0)$ has signature exactly $(0, 78, 0)$, verified from independently assembled structure constants.

[FACT, proven] A is a primitive idempotent (Peirce spectrum $\{0, \frac{1}{2}, 1\}$ at multiplicities $\{10, 16, 1\}$, both implementations). Its stabilizer within the electroweak $\mathfrak{su}(3)$ is exactly 4-dimensional: a 1-dimensional center $\mathfrak{u}(1)_Y$, commuting with color to 10^{-16} (hypercharge is color-blind), and a 3-

dimensional derived algebra $\mathfrak{su}(2)_{\text{weak}}$. A first candidate — the 3-dimensional centralizer of the full $\mathfrak{g}_2$ in $\mathfrak{f}_4$ — is disqualified by its own representation content (exclusively triplets and quintets on the 26, never doublets) and is recorded for completeness.

4. Three branching theorems and the exact charges

Theorem 1 (ratio 3). Under $\mathfrak{su}(3) \supset \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with $Y = \operatorname{diag}(a, a, -2a)$, the adjoint's doublets carry charge $\pm 3a$ against the fundamental doublet's a. Measured first ($0.343/0.1143 = 3.00$), proven afterward; the Standard Model's own lepton-to-quark doublet ratio $(-\frac{1}{2})/(\frac{1}{6}) = -3$ is this Lie-theoretic number.

Theorem 2 (8/1/7, with exact pre-registered predictions). $26 = (\mathbf{1}, \mathbf{8}) \oplus (\mathbf{3}, \mathbf{3}) \oplus (\bar{\mathbf{3}}, \bar{\mathbf{3}})$ — each pair being (color rep, electroweak-$\mathfrak{su}(3)$ rep), so $(\mathbf{3}, \mathbf{3})$ is a color triplet in the electroweak *fundamental*, not a weak triplet — forces exactly 8 doublets, 1 triplet, 7 singlets, and predicts — stated in exact form before the computation ran — one color-singlet at $Y = 0$, colored singlets at ratio -2, and a state at ratio $+4$ *forbidden*. All confirmed, including the required absence.

Theorem 3 (Peirce grade is weak type). By F_4-transitivity on primitive idempotents take $A = E_{33}$ WLOG (the statement is conjugation-invariant; the paper's own mediator-tied A is conjugate to it). Peirce grade counts indices equal to 3; the weak $\mathfrak{su}(2)$ rotates indices $\{1, 2\}$; a state is a doublet iff it carries exactly one such index — verbatim the Peirce-$\frac{1}{2}$ condition. Verified three independent ways (identical 16-dimensional subspaces by rank).

[FACT, verified twice] The census: 6 colored-doublet pairs, 2 lepton-doublet pairs (ratio ± 3), 3 colored-singlet pairs (ratio ∓ 2), one sterile singlet (0), one neutral triplet — 26 states, every multiplicity and ratio a branching consequence. With T_3 read at $\pm\frac{1}{2}$ on doublets (definitional) and the single calibration $Y(\text{colored doublet}) = \frac{1}{6}$, the charge $Q := T_3 + Y$ lands every state on a Standard Model value, $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$, the neutral triplet's own $\{0, \pm 1\}$ on the W-boson pattern consistent with its gauge-adjoint reading. Conjugate doubling throughout is the real-representation necessity: each entry is a state plus its mirror.

5. Chirality from $\mathbb{S}$'s own odd half

[FACT, proven] $\mathbb{O}e_8$ carries, under $\mathbb{S}$'s own multiplication, a uniform negative-definite Clifford structure: $\{(0, x), (0, y)\} = -2\langle x, y\rangle \cdot 1$ across **all eight** directions, the real direction included ($(0, 1)^2 = -1$; defect exactly 0 over all 64 pairs, both implementations). $\mathbb{S}$ supplies $\mathrm{Cl}(0, 8)$.

[FACT, proven] Seven generators give a central volume element (no chirality — odd Clifford algebras structurally cannot supply one). Eight give a genuine chirality operator: $\Omega^2 = +1$, eigenvalues ± 1 at multiplicity $8/8$, every generator swapping the halves exactly. **The mechanism is evenness, not signature**: any even Clifford algebra's volume anticommutes with its generators. What $(0, 1)$ specifically supplies is the completion of the odd seven to an even eight — and it is the unique direction all of $\mathfrak{g}_2$ fixes ($D(1) = 0$ for every derivation), which is why the lifted $\mathfrak{g}_2$, and color within it, provably never mix the halves.

[FACT, proven] Each half carries the 8-dimensional spinor of $\mathfrak{spin}(7)$ — the representation whose spinor-stabilizer is G_2 itself. The full even symmetry is $\mathfrak{so}(8)$ (28-dimensional, closure exact), and the two halves plus the vector representation are **pairwise inequivalent while sharing every**

quadratic invariant: $\dim \mathrm{Hom} = 0$ in all three tests, structure constants matched across representations at 10^{-16} — the complete, textbook triality, verified in two unrelated constructions (Pauli-tensor and octonion-multiplication models). Color's content on the two halves is identical — the strong force comes out vectorlike, parity-conserving, as observed.

6. The bridge theorem

[FACT, proven] $\mathrm{Stab}(A) \subset \mathfrak{f}_4$ is exactly 36-dimensional ($= \dim \mathrm{Spin}(9)$), contains the weak $\mathfrak{u}(2)$ (one line: commuting with L_A forces $D(A) = 0$), and acts irreducibly on $J_{\frac{1}{2}}$ (Schur commutant exactly 1; Casimir uniform at $|9/14|$ in intrinsic Killing normalization).

[FACT, proven] The nine traceless Peirce-0 directions satisfy a native Clifford relation on $J_{\frac{1}{2}}$: $\{L_u, L_v\}|_{J_{1/2}} = \frac{1}{4}\langle u, v\rangle\,\mathrm{Id}$, exhaustively exact. Within the slot family, the analogues of $\mathbb{O}e_8$'s $(0, 1)$ and mediator are identified by the identical invariant on both sides — full-$\mathfrak{g}_2$-fixedness picks r (fixed exactly) against m (moved): the same criterion that distinguishes $(0, 1)$ there. The two chiral columns are the sub-idempotent $\frac{1}{2}$-eigenspaces (each 8-dimensional, coinciding with the ninth generator's $\pm$ eigenspaces by rank); all eight slot generators are purely column-swapping.

[THM, the bridge] Under the dictionary $\varphi : \mathbb{O} \to \text{slot}$, $\varphi(x) \leftrightarrow (0, x)$: the composite J ($4L_m L_r$ in raw slot-unit normalization; 8 for unit-normalized generators — the coefficient is convention-dependent, both stated) is a genuine complex structure commuting with color; the color-centralizer of the gauge-side even algebra is exactly 2-dimensional (Schur-forced by the identical argument on both sides) and equals $\mathrm{span}\{J,\ \text{the pairing sum}\}$, mapping to $\{\Gamma_{\text{med}}\Gamma_{\text{real}},\ J_{\text{part}}\}$ **exactly** — the pair forced earlier by a one-sided computation, reappearing as a consequence of the dictionary. Because the gauge side's Clifford constant is positive while $\mathbb{O}e_8$'s is negative, the proven statement is: **same module, same even structure, opposite real signature.** Every load-bearing piece — the complex structure, the $\mathfrak{so}(8)$ action, the forced centralizer, $\mathfrak{g}_2$-compatibility — lives in the signature-insensitive even subalgebra and matches exactly; the single residue, the real quadratic form's sign, is confined to the odd generators' squares and stated rather than absorbed.

7. Weak isospin *is* the transported chirality: $T_3 = \lambda J$ and the quaternionic module

[FACT, proven] The prerequisite identity $\Gamma_9 \propto \Omega_8$ holds exactly: the ninth generator's action is the ordered product of the eight — so the columns are the genuine transported chirality split. Hypercharge preserves both columns at 10^{-16}; the raising and lowering generators, constructed as the Killing-orthogonal complement of the Cartan (never by search), are purely column-mixing, exactly.

[THM] $T_3 = \lambda J$ **identically** on $J_{\frac{1}{2}}$ ($\|T_3 - \lambda J\| \sim 10^{-16}$; $\lambda^2 = \frac{1}{24}$ exactly in the aligned normalization, $\lambda = 1/(2\sqrt{6})$ to 13 digits): weak isospin's own Cartan generator is a multiple of the bridge's complex structure — $J_{\text{weak}} = J_{\text{bridge}} = 4L_m L_r$, one operator reached independently from the electroweak stabilizer chain and from the $\mathbb{O}e_8$ dictionary. T_3 has, moreover, a closed form: $T_3 = 2\lambda[L_m, L_r]$, the idempotent frame's own inner derivation (residual 8×10^{-16}; equal to the annihilator-of-$(E_1' - E_2')$ extraction, whose defining property supplies the independent anchor).

[THM] $J_{\frac{1}{2}}$ is a **quaternionic module**, $\mathbb{H}^4$, with $\mathfrak{su}(2)_{\text{weak}}$ acting as $\mathfrak{sp}(1)$: $T_{\pm}^2 = -\mu^2\,\mathrm{Id}$,

$\{T_+, T_-\} = \{J, T_\pm\} = 0$, and $[T_+, T_-] = (2\mu^2/\lambda)\, T_3$ as an exact identity in each implementation's own normalization (the two pipelines' differing μ^2 values, $\frac{1}{26}$ and $\frac{1}{24}$, both satisfy the identity — the mismatch is proven cosmetic). The transported chirality operator is the quaternionic i; column-mixing by $T_\pm$ is quaternion multiplication.

The mechanical fact. Weak isospin acts *across* the transported chirality split; hypercharge acts *along* it. The interpretation of this fact is deliberately held open in §11.

8. The rank-6 torus: every generator named

[FACT, proven] $i \cdot L_{A_0}$ ($A_0 = A - \frac{1}{3}\mathbb{1}$) commutes with the entire SM chain (color, T_3, Y; all $\sim 10^{-16}$). The centralizer of the SM Cartan within $i\,L(J_0)$ is exactly 2-dimensional; the second direction is $i \cdot L_{E_1' - E_2'}$ (rank containment against the genuine Cartan), the sub-idempotent difference — which is also the ninth Clifford generator of §6, one object in two jobs. The complete rank-6 torus of $\mathfrak{e}_6$: two color Cartans; $T_3 = 2\lambda[L_m, L_r]$; Y (the stabilizer's center); $i \cdot L_{A_0}$; $i \cdot L_{E_1' - E_2'}$. **The two extra directions are the traceless diagonal multiplications of the complete primitive idempotent frame $\{A, E_1', E_2'\}$.** A bonus identity: the census's sterile singlet *is* A_0 — the unique color-and-weak singlet and the idempotent's traceless direction are one object.

9. The 27-census and the lattice point

[THM, pairing] On $\mathbb{C}^{27}$ (the census space — the new charges do not preserve the traceless 26), the six charges are simultaneously diagonalizable with **27 distinct weight vectors**: twelve conjugate pairs on which the $\mathfrak{p}$-side charges (Q_A, Q_w) are *equal* and the $\mathfrak{k}$-side charges (color, T_3, Y) are *negated* — a theorem of the symmetric/antisymmetric split, verified for all twelve — plus three self-conjugate states, which are **the idempotent frame itself**: A at $(Q_A, Q_w) = (\frac{2}{3}, 0)$, E_1', E_2' at $(-\frac{1}{3}, \pm 1)$, verified by joint eigenvalue equations. Q_A's spectrum is the Peirce weights; the 27 contains its own measuring frame as three of its states, and the former $26 \oplus 1$ split does not survive the torus.

[FACT, proven, exact arithmetic] For $Y' = \alpha Y + \beta Q_A + \gamma Q_w$, all quantities in $\frac{1}{6}$-units for Y and Q_A (Q_w at its natural $\pm\frac{1}{2}$), the defining equations are stated with their numeric coefficients so the convention travels inside them: the d_R-type pair carries $(Y_r, Q_{A,r}) = (\mp 2, -2)$, so holding it at -2 while its partner reaches the u_R slot $+4$ reads $-2\alpha - 2\beta = -2$ and $+2\alpha - 2\beta = +4$, forcing $\alpha = \frac{3}{2}$, $\beta = -\frac{1}{2}$ exactly. (In the mixed convention with Q_A raw, the same physics gives $\beta = -3$; the invariant is the contribution $\beta\, Q_A(d_R) = +1$, identical in both.) Downstream, in the same units: the lepton doublet carries $(Y_r, Q_{A,r}) = (-3, 1)$, so under the quark anchor $Y'(L) = \frac{3}{2}(-3) - \frac{1}{2}(1) = -5$. Without the u_R demand, the system holding quarks, leptons, and d_R at Standard Model values has the **unique** solution $(\alpha, \beta, \gamma) = (1, 0, 0)$ — the original hypercharge, where u_R was absent. At the u_R-forcing point, γ splits by anchor: the quark sector demands $\gamma = 0$ (chirality-blind), the lepton sector $\gamma = \pm 4$ (chirality-sensitive per column); jointly over-determined. **u_R's price**: whichever sector is not held fixed is displaced by exactly **two ratio units** (leptons to -5 as computed above, or quark columns to $\frac{3}{2}(1) - \frac{1}{2}(1) \pm 4 \cdot \frac{1}{2} = 1 \pm 2 = \{3, -1\}$), symmetric and exact. And e_R's slot is structurally empty: the census contains **no color-singlet weak-singlet pair at all** — the twelfth pair is the triplet's charged states.

10. The inequivalence theorem

[THM] The embedding $G_{SM} \hookrightarrow E_6$ selected by this construction is inequivalent to the $SO(10)$-GUT embedding. Witnesses, all computed: (i) this construction's 27 carries a weak-**triplet** matter state; the GUT 27 has none; (ii) it carries six colored weak-doublet pairs (three colors $\times$ two chiralities) where the GUT 27 carries three; (iii) the sector trade-off of §9, which the GUT embedding does not have. Internalized (Stage G): the centralizer of $i \cdot L_{A_0}$ in compact $\mathfrak{e}_6$ is exactly $46 = 36 + 9 + 1 = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$; all 46 generators are Peirce-block-diagonal at 10^{-16}, so **the** $\mathfrak{so}(10)$ **spinor sixteen and the Peirce-**$\frac{1}{2}$ **sixteen are the same subspace** in this construction (resolving the standing two-sixteens caution; the GUT route's 16 belongs to a different conjugate $\mathfrak{so}(10)$); and the corpus's weak $\mathfrak{su}(2)$, sitting inside this $\mathfrak{so}(10)$, branches the blocks as $10 \to 3 \oplus 7 \cdot 1$ and $16 \to 8 \cdot 2$ (Killing-normalized Casimir; triplet/doublet ratio $8/3$ exact) against the GUT class's hand-derived $10 \to (2, 2) \oplus 6 \cdot 1$ and $16 \to 4 \cdot 2 \oplus 8 \cdot 1$: **distinct branching multisets in both representations, hence distinct conjugacy classes of** $\mathfrak{su}(2)$ **inside one and the same** $\mathfrak{so}(10)$**.** The sole external input is the two-line textbook branching of the GUT class (Slansky).

Consequently: the absence of e_R, the price of u_R, and the presence of the triplet are not failures of this construction to reach the Standard Model — they are exact features of the embedding its own objects canonically select, and the precise location at which the published program's complexified and bi-Cayley extensions become necessary.

11. Discussion and scope

Relation to the published program. The destination — G_{SM} from $J_3(\mathbb{O})$ — is Dubois-Violette–Todorov's, in Boyle's idempotent-stabilizer form, generalized by Baez–Schwahn. What is this corpus's own: (i) a derivation, not an assumption, of *why octonions* — the enumeration theorem proves the $1+3+3$ non-associative grading forced; (ii) the mediator as the single origin of both color and complexification; (iii) the bridge and $T_3 = \lambda J$ theorems connecting the gauge structure to a chirality mechanism the program does not have; (iv) the inequivalence theorem and the boundary results of §§9–10, which sharpen the program's own stated limits into exact statements.

The fork, held open by design. The transported chirality grading turns out to be $\mathfrak{su}(2)_{\text{weak}}$'s own Cartan axis — an object with no Lorentz pedigree. *Reading A (transmuted)*: the weak sector does not couple to a chirality; it contains one as an axis of its own quaternionic action. *Reading B*: the physical γ^5-analogue is a different object, living, if anywhere in this corpus, at the fermionic state level where Lorentz structure enters. The γ-ambiguity of §9 is prior to this question, not a resolution of it. Nothing in this paper decides the fork, and the two readings are recorded at equal rank with B's evidence currently heavier.

Open, named. Canonical selection: whether the construction's own objects select a γ-anchor (and hence a chirality-sensitivity) rather than leaving it external. The electroweak mixing angle: a reason for the Weinberg combination beyond $Q = T_3 + Y$ landing correctly. The fermionic program (quasi-free states on the $\mathrm{Cl}(0, 8)$ module, a state-dependent chirality-odd antiunitary, quantization of this census): the named frontier, deliberately not begun here.

What was not attempted. Gauge dynamics, a Lagrangian, the Higgs mechanism, generations, or any claim beyond the group-, representation-, and module-theoretic content above.

Verification statement. Two independent implementations (different octonion arithmetic, different $J_3(\mathbb{O})$ representations, different extraction routes; the Clifford side built by Pauli-tensor and octonion-multiplication models respectively), agreeing on every invariant checkpoint — dimensions, ranks, exact fractions, ratios, Hom-dimensions, branching multisets — with all convention-dependent scalars reconciled by stated unit systems, and the lattice results in exact rational arithmetic. Pre-registered expectation lists preceded every computation stage on both sides; the complete checkpoint tables, the deviations ledger, and the process history are in the development notes.

References

- Companion papers of this corpus: the enumeration paper (*The Non-Associativity Selector*); $J_3(\mathbb{O})$*: Construction, the Peirce Split, and Verified Frame Relativity via* F_4; *From* $J_3(\mathbb{O})$ *to* $E_7(-25)$; the Foundation and Keystone papers for $(\mathbb{S}, \tau)$, the mediator, and the quantization chain.
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Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.